

AI for Green Transition

Understanding Carbon Footprints, Green Technologies & Building an AI-Powered Sustainable Solution

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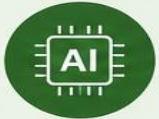
Why Sustainability Matters



Climate change is accelerating.



Global carbon emissions continue to rise.



AI and digital technologies consume increasing energy.



Green technologies help reduce environmental impact.



Every individual and organization has a role in sustainability.

“You can’t improve what you don’t **measure.”**



| Decoding Carbon & Tech Footprints

How Digital Technology Impacts the Environment

- Data centers are now the engine of the global economy, but they are incredibly power-hungry. As we consume streaming media, run high-definition video tools, and integrate generative AI pipelines, the energy and water footprints of these facilities skyrocket.
- Tools like global CO₂ models (MIT En-ROADS) show that without decarbonizing the basic grid, increased digitization risks outpacing our climate mitigation targets.



| Decoding Carbon & Tech Footprints

The Precision of Carbon Math

- Ecological Footprint and EcoTransit calculators allow businesses to quantify hidden tailpipe and logistic emissions. Understanding the baseline is the primary step of real sustainability.
- By mapping out complete transits and everyday resource cycles, we transition from vague carbon awareness to highly localized, target-driven climate action plans.

Sustainability Tools Explored



Ecological Footprint Calculator

Measures resource consumption and the demand on Earth's ecosystems.



EcoTransit

Calculates carbon emissions generated from different modes of transportation and logistics operations.



MIT En-ROADS

An interactive climate simulation model that demonstrates how different policies and technologies affect global CO₂ emissions and temperature.



NANDO AI

Uses Artificial Intelligence, Computer Vision, and IoT to improve waste sorting, increase recycling efficiency, and reduce landfill waste.



e-Amrit

Estimates CO₂ emissions and compares Internal Combustion Engine (ICE) vehicles with Electric Vehicles (EVs) to promote cleaner mobility.



SCGJ (Skill Council for Green Jobs)

Develops industry-ready green skills through training and certification in renewable energy, EV technology, waste management, and sustainability.

RESULTS

Your personal Earth Overshoot Day is:

16. Jun 

If everyone lived like you, we would need

2.2 Earths 





Electric vehicle 1

Two-Whe ▾

Ola Electr ▾

Ola S1 Pr ▾

Electric vehicle 2

Four-Whe ▾

Tata ▾

Tata Tigor ▾

Electric vehicle 3

Two-Whe ▾

Revolt Bik ▾

Revolt Bik ▾

Search

Showing Result



Make	Ola Electric
Model	Ola S1 Pro
Ex Showroom Price	₹ 100000
Battery Size (kWh)	4
Battery Type	Lithium Ion
Charger Rating(kW)	4
Driving Range(km)	120



Make	Tata
Model	Tata Tigor
Ex Showroom Price	₹ 1250000
Battery Size (kWh)	26
Battery Type	Lithium Ion
Charger Rating(kW)	25
Driving Range(km)	306



Make	Revolt Bike
Model	Revolt Bike
Ex Showroom Price	₹ 120000
Battery Size (kWh)	3
Battery Type	Lithium ion
Charger Rating(kW)	3
Driving Range(km)	150

Harnessing AI & Green Talent

NANDO AI Smart Waste Scanner

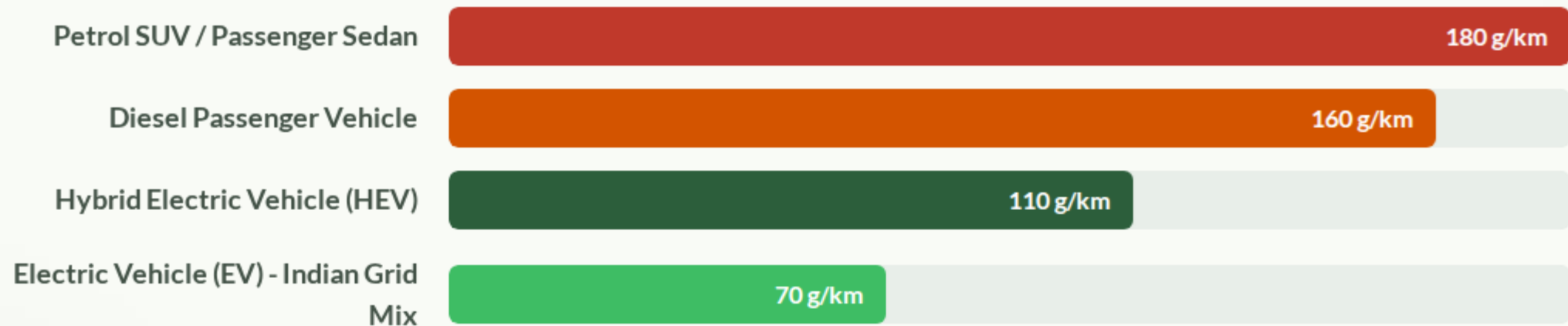
Using machine learning, computer vision, and IoT sensors, NANDO scans waste in real time to prevent sorting contamination. Businesses implementing this tool have boosted recycling yields by up to 60%, drastically reducing local landfill emissions.

The Skill Council for Green Jobs (SCGJ)

Technology alone is insufficient. SCGJ builds the critical workforce pipeline by certifying professionals in solar power (Suryamitra), wind installations, electric vehicle maintenance, and localized waste management systems.



Emission Abatement: EVs vs ICE



Based on NITI Aayog's e-Amrit CO2 Calculator: Switching to an EV reduces tailpipe emissions to zero and total lifecycle emissions by over 60%, even when utilizing India's current thermal-heavy electricity grid.

| The Proposed Solution: Eco-Hub



AI Waste Audit

Run a localized pilot of NANDO-inspired smart waste cameras above sorting stations to track and gamify recycling targets for users.



E-Amrit Integration

Deploy localized EV transport calculators for commuters, illustrating the direct cost and carbon savings of shifting to electric mobility.



SCGJ Green Skills

Sponsor student/employee green-skills training, creating certified sustainability ambassadors capable of handling on-site solar installations.



EcoHub AI

AI-Powered Carbon Footprint & Sustainability Assistant



Project Objective

Develop a Python-based application that helps users understand and reduce their environmental impact using Artificial Intelligence.



Features



Carbon Footprint Calculator



AI Sustainability Assistant



Personalized Eco-Friendly Recommendations



Green Score Dashboard



Progress Tracking



Daily Sustainability Tips



Technology



Python



Streamlit



Pandas



Gemini/
OpenAI API



Plotly



Expected Outcome

Help users make informed decisions to reduce carbon emissions and promote sustainable living.



Understand your impact



AI-powered insights



Track your progress



Adopt better habits



Build a sustainable future

Project Implementation Roadmap

Phase 2: Prototyping

Integrate low-cost smart waste sensors and build the gamified recycling leaderboard interface.

Phase 4: Training & Scale

Partner with certified trainers to provide accredited SCGJ Solar / EV basic technician credentials.

Phase 1: Audit

Setup ecological footprint calculators to baseline existing power, transport, and waste patterns.

Phase 3: EV Charger

Install basic EV charging infrastructure supported by real-time emission savings trackers.

Thank You

Small Actions + Smart Technology = Sustainable Future